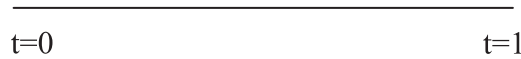


- 3) Take first order condition and solve for W_{t+1}^* (* for optimal)
- 4) Set our initial guess equal to our Bellman equation evaluated at its maximum W_{t+1}^*
- 5) Identity the unknown coefficient from our guess
- 6) Use the coefficient to identity our policy function from W_{t+1}^*

Let us now look at the solution. Consider the diagram below



At $t=0$ $C_t=0$ because the muncher just starts eating at $t=1$ the consumption is recorded so consumption begins effectively at $t=1$. Similarly W_0 is the initial cake size which is fixed for the model. In other words, we need to maximize W_{t+1} for each time interval to maximize the consumption

We know $C_t = W_t - W_{t+1}$

C_t is maximized to W_{t+1} is maximized as W_t is fixed. So, we can rewrite the question as under

$$\max_{\{W_{t+1}\}_{t=1}^T} \sum_{t=0}^T \ln(W_t - W_{t+1})$$

Hence our bellman equation is

$$V(W_t) = \max_{\{W_{t+1}\}} \{\ln(W_t - W_{t+1}) + \beta V(W_{t+1})\} \quad \dots\dots(22.39)$$

We make a guess of $V(W_t)$

Let that be $V(W_t) = C_1 + C_2 \ln W_t$

Moving one period ahead $V(W_{t+1}) = C_1 + C_2 \ln W_{t+1}$

Substituting this result into our bellman equation we have

$$V(W_t) = \max_{W_{t+1}} \{\ln(W_t - W_{t+1}) + \beta [C_1 + C_2 \ln W_{t+1}]\}$$

Take F O C of above equation for W_{t+1}^*

$$\frac{\partial V(W_t)}{\partial W_{t+1}} = \frac{-1}{W_t - W_{t+1}} + \frac{\beta C_2}{W_{t+1}} = 0$$

$$W_{t+1}^* = \frac{\beta C_2 W_t}{1 + \beta C_2} \quad \dots\dots\dots(22.40)$$

Set our initial guess equal to our bellman equation evaluated W_{t+1}^*

$$C_1 + C_2 \ln W_t = \ln \left(W_t - \frac{\beta C_2 W_t}{1 + \beta C_2} \right) + \beta \left[C_1 + C_2 \ln \left(\frac{\beta C_2 W_t}{1 + \beta C_2} \right) \right]$$

$$\begin{aligned}
&= \ln\left(\frac{W_t}{1+\beta C_2}\right) + \beta \left[C_1 + C_2 \ln\left(\frac{\beta C_2 W_t}{1+\beta C_2}\right) \right] \\
&= \ln(W_t) - \ln(1+\beta C_2) + \beta \{ C_1 + C_2 [\ln(\beta C_2) + \ln(W_t) - \ln(1+\beta C_2)] \} \Rightarrow C_1 + C_2 \ln W_t \\
&= \underbrace{\beta C_1 - \ln(1+\beta C_2) + C_2 [\ln(\beta C_2) - \ln(1+\beta C_2)]}_1 + \underbrace{(1+\beta C_2) \ln(W_t)}_2
\end{aligned}$$

We can see that our constant term in our guess C_1 must be equal to the constant terms on right hand side of our boxed in equation denoted by (1). We require however that the coefficient on $\ln(W_t)$ need to be equivalent on both sides

This implies $C_2 = 1 + \beta C_2$

$$C_2^* = \frac{1}{1-\beta} \quad \dots\dots(22.41)$$

Use the coefficient to identity our policy function from W_{t+1}^*

Recall from equation (22.40) our optimal value

$$\text{Let } g(W_t) = W_{t+1}^* = \frac{\beta C_2^* W_t}{1+\beta C_2} \quad \dots\dots(22.42)$$

Plugging equation 22.41 in equation 22.42 in above equation

$$W_{t+1}^* = \frac{\beta \left(\frac{1}{1-\beta}\right) W_t}{1 + \beta \left(\frac{1}{1-\beta}\right)}$$

Further rearranging we obtain our policy function for how much cake should be left for each period

$$g(W_t) = \beta W_t = W_{t+1}^*$$

The corresponding consumption policy is denoted by

$$\begin{aligned}
C_t^* &= W_t - W_{t+1}^* \\
&= W_t - g(W_t) = W_t - \beta W_t = (1 - \beta)W_t
\end{aligned}$$

Therefore, our optimal consumption function is

$$C_t^* = (1 - \beta)W_t \quad \dots\dots\dots(22.43)$$

22.11 LET US SUM UP

We initiated our discussion with clarifications about distinctions between (i) static & dynamic optimization approaches, and (ii) function and functional. Then we enumerated various dynamic optimization approaches, discussed the essential idea of each approach, stated the theoretical results which form the foundations of the approaches, and then explained how to use each one for solving dynamic optimisation problems. More explicitly, first we discussed the Calculus of Variations approach, stated the foundational theorem justifying the basis of the approach, and then explained how to use it in solving appropriate problems. Next, we explained the dynamic programming approach for dynamic

optimization, and used it in deriving optimality criteria of discrete dynamic optimisation. Finally, we explained how to use the approach in solving relevant problems.

22.12 KEY WORDS

Euler-Lagrange equation	:	The necessary condition for optimisation in calculation of variation which can be found by setting 1 st variation of functional to zero. Necessary and sufficient condition for optimal: 2 nd variation of functional is positive definite for maximum and negative definite for maximum.
Function	:	The function of with its domain as a set of functions.
Objective functional	:	The function which needs to be optimized i.e., either maximized or minimized.
Isoperimetric Problems		Types of problems for which we seek optimal value of functions which are subjected to integral constraint, in addition to ‘ordinary’ boundary conditions. Such problems are called isoperimetric problems
Bellman-equation	:	Value functional of objective functional. $V(.) = \max_u \{f(x, u) + \beta V(g(u, x))\}$

22.13 SOME USEFUL BOOKS

Chiang.C.Alpha, 1992, *Elements of Dynamic Optimisation, International Editions*, Mc Graw – Hill, Inc.

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Seierstad Atle, Sydsaeter Kunt, 1987, *Optimal Control Theory with Economic Applications, First Edition Elsevier Science B.V.*, The Netherlands

Stokey.L.Nancey, Lucas.E.Robert, Jr., Prescott.C.Edward, 1989, *Recursive methods in economic dynamics*, first edition, Harvard University press.

Weber.A.Thomas, 2011 *Optimal Control Theory With Applications In Economics*, The MIT press.

Whittle. Peter, 1996, *Optimal Control, Basic and Beyond* John Wiley& Sons Ltd.

22.13 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

1) Problem 1 $J(y) = \int_0^1 (y'^2 + y^2 + 4ye^t) dt$

We know from E – L condition that

$$\frac{\partial v(.)}{\partial y(t)} - \frac{d}{dt} \frac{\partial v(.)}{\partial y'(t)} = 0 \quad \text{where } v(.) = y'^2 + y^2 + 4ye^t$$

hence $\frac{\partial v(.)}{\partial y(t)} = 2y + 4e^t, \quad \frac{\partial v(.)}{\partial y'(t)} = 2y'(t)$

Therefore $2y + 4e^t = 2y''(t)$

$$\Rightarrow y''(t) - y(t) = 2e^t$$

First we find particular integral y_p

Let $y_t = Bte^t \Rightarrow y'_t = B[te^t + e^t]$

$$y''(t) = B[te^t + e^t]' = B[te^t + e^t + e^t] = Be^t + 2Be^t$$

So $Bte^t + 2Be^t - Bte^t = 2e^t$

$$\Rightarrow 2Be^t = 2e^t \Rightarrow B = 1 \quad \text{so } y_p = 1 \cdot te^t$$

for complimentary function y_c let $y(t) = Ae^{rt}$

then $y'' = Ar^2e^{rt}$

$$\Rightarrow Ar^2e^{rt} - Ae^{rt} = 0$$

$$\Rightarrow r^2 - 1 = 0 \Rightarrow r^2 = 1 \Rightarrow r = \pm 1$$

So $y_c = A_1e^{r_1t} + A_2e^{r_2t} = A_1e^{-t} + A_2e^{+t}$

We know $y(0) = 0 \Rightarrow A_1 + A_2 = 0 \Rightarrow A_1 = -A_2$

We know $y(1) = 1 \Rightarrow A_1e + A_2e^{-1} = 1$

$$\Rightarrow -A_2e + A_2e^{-1} = 1 \Rightarrow A_2 = \frac{e}{1-e^2}$$

Using boundary condition

$$y(t) = y_c + y_p = \frac{e}{e^2 - 1} e^t + \frac{e}{1 - e^2} e^{-t} + te^t \text{ (Ans)}$$

Check Your Progress 2

1) Optimise $\int_0^1 \{[x'(t)]^2 + 10x(t) \cdot t\} dt$

Here $v = x'^2 + 10x \cdot t$ given $x(0) = 2$ $x(1) = 3$

We know $\frac{\partial v(.)}{\partial x} - \frac{d}{dt} \frac{\partial v(.)}{\partial x'} = 0$

$$\Rightarrow 10t - \frac{d}{dt} 2x' = 0$$

$$\Rightarrow 5t - x'' = 0$$

This can be written $\frac{d^2x}{dt^2} = 5t$

Let $\frac{dx}{dt} = y \Rightarrow \frac{dy}{dt} = 5t$

$$\Rightarrow dy = 5t dt$$

$$\Rightarrow \int dy = \int 5t dt$$

$$\Rightarrow y = 5 \frac{t^2}{2} + c_1$$

$$\Rightarrow \frac{dx}{dt} = 5 \frac{t^2}{2} + c_1$$

$$\Rightarrow dx = 5 \frac{t^2}{2} dt + c_1 dt$$

$$\Rightarrow \int dx = \int 5 \frac{t^2}{2} dt + \int c_1 dt$$

$$\Rightarrow x = \frac{5}{2} \frac{t^3}{3} + c_1 t + c_2$$

Using boundary condition

$$x(t_0) = 2 \Rightarrow 2 = 0 + 0 + c_2 \Rightarrow c_2 = 2$$

$$\text{Similarly } x(1) = 3 \Rightarrow 3 = \frac{5}{2} \frac{1}{3} + c_1 + 2$$

$$\Rightarrow c_1 = 1/6$$

Therefore $x^*(t) = \frac{5}{6} t^3 + \frac{1}{6} t + 2$ (Ans)

2) Let $V = (x(t), \dot{x}(t), t) = 2x'^2(t) + 24x(t) \cdot t$

Writing Euler-Lagrange equation

$$\frac{\partial V(.)}{\partial x(t)} - \frac{d}{dt} \left[\frac{\partial V(.)}{\partial x'(t)} \right] = 0$$

$$\Rightarrow 24t - \frac{d}{dt}(4x'(t)) = 0$$

$$\Rightarrow 24t - 4x''(t) = 0$$

$$\Rightarrow x''(t) = 6t$$

$$\Rightarrow \frac{d}{dt}x'(t) = 6t$$

$$\Rightarrow \int dx'(t) = \int 6t dt$$

$$\Rightarrow x'(t) = 6\frac{t^2}{2} + C_1 \text{ where } C_1 \text{ is constant}$$

$$\Rightarrow \frac{d}{dt}x(t) = 6\frac{t^2}{2} + C_1$$

$$\Rightarrow \int dx(t) = \int (3t^2 + (C_1))dt$$

$$\Rightarrow x(t) = t^3 + C_1t + C_2 \text{ where } C_2 \text{ is constant}$$

We need to find C_1, C_2 for which boundary condition can be used.

$$x(t_0) = x_0 = 0 = C_1 t_0 + C_2 \Rightarrow C_2 = 0$$

$$\text{at } t = t_f \quad x(t_f) = x_f = 2 \quad \Rightarrow \quad 8 + 2C_1 + C_2 = 2$$

$$\Rightarrow C_1 = -3$$

Hence, we can write $x(t) = -3t + t^3$ This equation satisfied initial and boundary value constraint.

In order to verify we use Legendre condition we calculate $\frac{\partial^2 V(\cdot)}{\partial x'^2(t)} = \frac{\partial}{\partial x'(t)} 4x'(t) = 4 > 0$ hence the value of $x^*(t)$ minimizes the functional $J(\cdot)$

22.14 EXERCISES

Q 1. Solve Euler-Lagrange equation for

$$J() = \int_0^1 (x'^2 - 2xx' + 10tx) dt \text{ subject to } x(0)=0 \text{ and } x(1) = 2$$

Ans. We have to find E.L. equation for

$$J(\cdot) = \int_0^1 (x'^2 - 2xx' + 10xt) dt \quad \text{s.t. } x(0) = 0 \quad x(1) = 2$$

$$\text{So here } v() = x'^2 - 2xx' + 10xt$$

$$\text{As per theorem } \frac{\partial v(\cdot)}{\partial x} - \frac{d}{dt} \left[\frac{\partial v(\cdot)}{\partial x'(t)} \right] = 0 \quad \dots (a)$$

$$\frac{\partial v(\cdot)}{\partial x} = -2x' + 10t$$

$$\frac{\partial v}{\partial x'} = 2x' - 2x \quad \text{so } \frac{\partial}{\partial t} \left[\frac{\partial v(\cdot)}{\partial x'} \right] = \frac{d}{dt} [2x' - 2x] = 2x'' - 2x'$$

Putting the results in equation (a)

$$-2x' + 10t - 2x'' + 2x' = 0$$

$$\Rightarrow 2x'' = 10t$$

$$\text{Let } x' = y \quad \text{so } 2y' = 10t$$

$$\Rightarrow \int 2dy = \int 10t dt$$

$$\Rightarrow y = 10 \frac{t^2}{2} + c_1$$

$$\Rightarrow \int dx = \int \left[10 \frac{t^2}{2} dt + c_1 dt \right]$$

$$\Rightarrow x(t) = \frac{10}{2} \cdot \frac{t^3}{3} + c_1 t + c_2$$

$$\text{at time } t = 0 \text{ we get } 0 = c_2 = 0$$

$$\text{at time } t = 1 \text{ we get } 2 = 5 \cdot \frac{1}{3} + c_1 \Rightarrow c_1 = \frac{1}{3}$$

$$\text{Therefore } x^*(t) = 5 \frac{t^3}{3} + \frac{1}{3} t \text{ (Ans)}$$

Q. 2 . Find the $x^*(t)$ which in turn shall minimize

$$\int_0^1 [x'(t)]^2 dt \text{ subject to } \int_0^1 x(t) dt = K, \quad x(0) = 1, \quad x(1) = 2$$

Ans. This is a case of isoperimetric problem the augmented objective function is

$$(x')^2 - \lambda x \text{ it's E - L equation is } -\lambda - 2x'' = 0$$

$$\Rightarrow 2x'' + \lambda = 0$$

$$\text{The solution of } x(t) \text{ is equal to } \frac{-\lambda t^2}{4} + c_1 t + c_2$$

$$\text{The constraint equation is } \int_0^1 x(t) dt = k$$

$$\Rightarrow \int_0^1 \left(-\frac{\lambda t^2}{4} + c_1 t + c_2 \right) dt = k \quad \dots (a)$$

$$x(0) = 1 \Rightarrow c_2 = 1$$

$$x(1) = 2 \Rightarrow \frac{-\lambda}{4} + c_1 + 1 = 2 \Rightarrow c_1 = 1 + \frac{\lambda}{4}$$

Using the value of c_1 and c_2 in equation (a)

$$\text{We get } \int_0^1 \left(-\frac{\lambda t^2}{4} + t + \frac{\lambda}{4} t + 1 \right) dt = k$$

$$\Rightarrow \frac{1}{4} \int_0^1 (-\lambda t^2 + 4t + \lambda t + 4) dt = k$$

$$\Rightarrow \left[-\lambda \frac{t^3}{3} + \frac{4t^2}{2} + \lambda \frac{t^2}{2} + 4t \right]_0^1 = 4k$$

$$\Rightarrow -\frac{\lambda}{3} + 2 + \frac{\lambda}{2} + 4 - 0 = 4k$$

$$\Rightarrow \frac{\lambda}{2} - \frac{\lambda}{3} = 4k - 6$$

$$\Rightarrow \frac{1}{6} \lambda = 4k - 6 \Rightarrow \lambda = 24(k - 1)$$

Using value of c_1, c_2 and λ

$$x^*(t) = -6(k-1)t^2 + t + 6(k-1)t + 1$$

Q3. Find the extremal for functional

$$J() = \int_{t=0}^{t_f} (2\dot{x}^2(t) + 2x(t) \cdot t) dt$$

$$t = t_0 = 0 \text{ such that } x(t_0) = 0, \quad x_{t_f} = 2$$

(Hint use transversality equation (22.26))

Ans. We have to find the extremal for the functional

$$J(.) = \int_{t_0=0}^{t_f} (2x'^2 + 24xt) dt$$

Where the left end point is fixed i.e., $x(t_0) = 0$ at $t_0 = 0$

We can assign that as point A. The right end point t_f is free but $x_{t_f} = 2$

$$v(x(t), x'(t), t) = 2x'^2 + 24xt$$

Writing Euler-Lagrange equation

$$= \frac{\partial v}{\partial x} - \frac{d}{dt} \left[\frac{\partial v(t)}{\partial x'} \right] = 0 \Rightarrow 24t - 4x''(t) = 0 \Rightarrow x''(t) = 6t.$$

Solving the differential equation $\ddot{x}(t) = 6t$ will give us

$$x(t) = t^3 + c_1 t + c_2 \quad \dots (1)$$

We want to find c_1 and c_2 so doing we use the boundary condition

$$x(t_0) = 0 = c_2 \Rightarrow c_2 = 0$$

$$\text{at } t = t_f, x_f = 2 = c_1 t_f + c_2 + t_f^3 \Rightarrow \frac{2 - t_f^3}{t_f} = c_1 \quad \dots (2)$$

because c_1 and c_2 value is known the description or trajectory is known provided the value of t_f is known.

Using transversality condition (when x_{tf} is fixed, t_f is free)

$$\left[v(t) - \frac{\partial v(t)}{\partial x'(t)} \cdot x'(t) \right] \Big|_{t=t_f} = 0$$

$$\Rightarrow [2x'^2 + 24xt - 4(x')^2] = 0 \Rightarrow 2x'^2 = 24x(t), t \quad \dots (1a)$$

$$x'(t_f) = 3t_f^2 + c_1 \text{ (by differentiating eq(1))} \quad \dots (3)$$

Using (3) in (1a) and using (2) in that and setting $x_{tf} = x_f = 2$

$$\text{We get } t_f^6 - 4t_f^3 + 1 = 0 \text{ let } z = t_f^3$$

$$\Rightarrow z^2 - 4z + 1 = 0 \Rightarrow z = 2 \pm \sqrt{3}$$

$$\text{So } z_1 = 2 + \sqrt{3} \text{ and } z_2 = 2 - \sqrt{3}, \quad z_1 = 3.732 \quad z_2 = 0.2679$$

$$\text{So } t_{f1} \text{ we get } c_1 = -1.117 \text{ and for } t_{f2} \text{ we get } c_1 = 2.6871$$

$$\text{So } t_{f1} = \sqrt[3]{3.732} \quad t_{f2} = \sqrt[3]{0.2679} = .6446$$

$$\text{for } t_{f1} \text{ we get } c_1 = -1.117 \text{ and for } t_{f2} \text{ we get } c_1 = 2.6871$$

$$\text{then } x(t) \text{ is } -1.117t + t^3 \rightarrow \text{one solution}$$

$$\text{or } 2.6871t + t^3 \text{ another solution}$$

$$\text{these above equations are a result of } x(t) = c_1t + t^3$$

Q4. Derive Bellman equation for $\max \sum_{t=0}^{t-T} f(u_t, x_t)$

$$\text{Subject to } u_t = x_t - x_{t+1}$$

Ans. In the space of the sequence $\{u_t, x_t\}_{t=0}^{\infty}$ such that $u_t \in P^m$ and $x_t \in P$ we have to choose a sequence $\{u_t^*, x_t^*\}_{t=0}^{\infty}$ that maximizes the sum

$$\max_{\{v\}} \sum_{t=0}^{\infty} \beta^t f(x_t, u_t)$$

Subject to the sequence of budget constraint

$$x_{t+1} = x_t - u_t, \quad t = 0, \dots, \infty$$

$$x_0 \text{ is given let } x_{t+1} = g(x_t, u_t)$$

$$\text{where } 0 < \beta \leq \frac{1}{1+\rho} < 1 \text{ where } \rho > 0$$

By applying the principle of dynamic programming the 1st order necessary condition for this problem is given by Bellman equation

$$v(x_t) = \max_{u_t} \{f(x_t, u_t) + \beta v(g(x_t, u_t))\}$$

which usually written as

$$v(x) = \max_u \{f(x, u) + \beta v(g(x, u))\} \text{ (Ans).}$$

UNIT 23 INTERTEMPORAL OPTIMISATION-II

Structure

- 23.0 Objectives
- 23.1 Introduction
 - 23.1.1 Summary of Optimization & Dynamic Optimization so far
- 23.2 First Order Necessary Condition: Euler Lagrange Equation
- 23.3 Hamiltonian Formation of Optimal Control Problem
- 23.4 Hamiltonian Optimality Condition in Special Cases
- 23.5 Transversality Condition
- 23.6 The Discounted Problem
- 23.7 Second Order Condition
- 23.8 Let Us Sum Up
- 23.9 Key Words
- 23.10 Answers/Hints to Check Your Progress Exercises
- 23.11 Exercises

23.0 OBJECTIVES

After going through this unit, you should be able to

- show how Hamiltonian formation is derived for constrained dynamic optimisation;
- determine necessary and sufficient condition for optimizing Hamiltonian functional; and
- elucidate two special types of dynamic optimisation techniques encountered in economics viz., infinity horizon models and optimisation models with discounting.

23.1 INTRODUCTION

In the previous unit, we have discussed the problem of optimisation of a functional under integration sign by using calculus of variations. Moreover, we have noted various end-point conditions with their transversality conditions. We derived necessary and sufficient conditions for extremum. Here we apply calculus of variations to find solutions to optimal control problems.

Before we go into details of '*Optimal Control Theory*', we summarise our discussion, so far, of optimisation, and in the process, introduce some concepts, essential for *any* further discussion on Optimisation.